

Fig. 6: GHDL

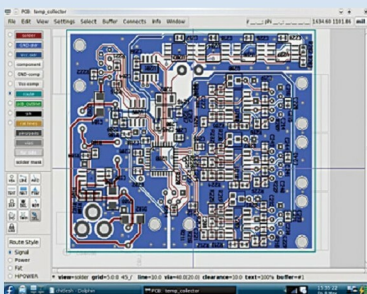


Fig. 7: PCB design

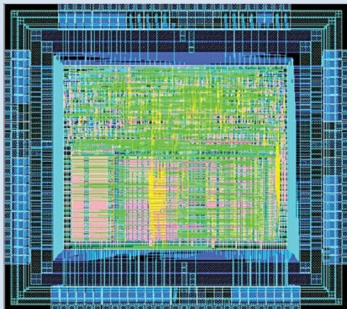


Fig. 6: An MIPS-R3000 of 52000 transistors that has been completely designed using Alliance CAD tools

tion tool helps designers to edit their schematics and to simulate them. This category includes both general-purpose drawing programs and specific computer aided design (CAD) tools, like Xcircuit and various simulators from general-purpose to mixed-level (for example, gnuap, ngspice or gspiceui). When design tools allow one to generate publishable quality of circuit diagrams, simulators perform all types of circuit analysis at a level beyond the capabilities of Spice implementations. These tools are multi-lingual, can mimic different variants of Spice and can also support new languages, such as Verilog-AMS.

an updated report of the violations in the design, if any. After the design is complete, a hierarchical circuit extractor re-extracts the part of the circuit that has been changed and permits an interactive stretching and compaction, routing the tools to work under and around existing connections.

Successful compilation and verification. FEL provides a hardware description language (HDL) simulation environment that allows you to verify functional timing models of the designs. It contains tools like GHDL, quite universal circuit simulator (Qucs), FreeHDL, Icarus

Design rule checks (DRC).

FEL platform allows layouts for ASICs using tools like Magic, Electric, Toped and Netgen. These also help in generating database formats, like GDSII stream and crystallographic information file (CIF), from a given layout to enable easy file transfer based on industry standards. During operation, a continuous DRC operates in the background and provides

Verilog, GTKWave and Drawtimg to scale-up the multiple levels of complexities and abstractions in the design. These support both VHDL and Verilog designs and allow cross-reference generation in HTML. Standard cell libraries help generate automatic layouts from VHDL description, which is then successfully compiled and run on DLX and LEON1 SPARC processor.

Logical design flow. Tools like pharosc, Alliance and gds2pov are more such design tools in FEL that not only route and automate the layout generation but also layout the design flow. These provide a complete real-time logic (RTL) for CIF and GDSII formats and create a three-dimensional view of the scenes described in GDSII files. Complete checking of the model and format proofing occurs at this stage.

Experience talks

Students working on Fedora have liked FEL a lot from the time it was just a project. They not only want to contribute towards it as users but also as developers.

A student at <http://blog.gmane.org/> says that she has been working on Fedora electronic spin since quite some time now. She has liked working with this so much that she has also worked as an ambassador promoting the advantages of FEL.